1 Conditional Downstream-Loss Bound and Exact Worst-Case Sensitivity Analysis

This supplementary appendix provides the complete derivation and computational details underlying the conditional downstream-loss analysis in the main manuscript. The analysis connects bounded operational predictor errors to downstream executable solution quality, characterizes the dependence of the resulting bound on problem size, and examines the most adverse discrete matching response that can be induced within a prescribed predictor-error set. Throughout this appendix, the exact pattern master and the A-CAM proxy master are defined over the same feasible pattern-selection set. If a restricted candidate-pattern set is used, all theoretical comparisons are interpreted relative to the exact pattern master defined over that same set. In the small-instance worst-case audit below, the complete feasible pattern and matching sets are enumerated.

1.1 Downstream-Loss Benchmark and Error Propagation

Let $\mathcal{B}$ denote the feasible vehicle–order pattern set and let $\mathcal{X}$ denote the corresponding feasible pattern-selection set. For each pattern $b \in \mathcal{B}$, let S_b denote the set of orders served by that pattern, and let v_b denote its exact executable profit obtained by fixing the vehicle–order pattern and solving the corresponding joint O-CAM route-and-pricing problem to optimality. For any feasible pattern selection $x \in \mathcal{X}$, define

$$V(x) = \sum_{b \in \mathcal{B}} v_b x_b. \tag{1}$$

Let

$$x^* \in \arg\max_{x \in \mathcal{X}} V(x), \qquad Z^* = V(x^*). \tag{2}$$

For a pattern selection $\widehat{x}$ returned by A-CAM, the downstream loss of interest is therefore

$$Z^* - V(\widehat{x}). \tag{3}$$

The loss in (3) can be nonzero because predictor-based pattern values may change the relative ranking of feasible matchings. For example, consider two mutually exclusive patterns b_1 and b_2 satisfying

$$v_{b_1} > v_{b_2}, \qquad \widehat{a}_{b_1} < \widehat{a}_{b_2},$$

where $\widehat{a}_b$ denotes the A-CAM proxy value of pattern b. The exact pattern master selects b_1, whereas A-CAM selects b_2. Under Policy 2 of the main manuscript, post-optimization reoptimizes routing and pricing conditional on the selected matching, but it cannot replace b_2 by b_1. Thus, predictor-induced ranking reversals can translate directly into downstream executable objective loss.

To characterize this loss, we separate three sources of error. First, let a_b^0 denote the A-CAM-style pattern value evaluated using the exact aggregate service attributes of pattern b, and define the structural aggregation error as

$$A_b = |v_b - a_b^0|. \tag{4}$$

This term can remain nonzero even when the operational predictors are exact because the aggregate A-CAM representation need not reproduce the full order-specific executable evaluation of the fixed-pattern O-CAM problem.

Second, suppose the pattern-level predictor errors satisfy

$$|\widehat{\ell}_b - \ell_b^*| \le \epsilon_\ell, \qquad |\widehat{w}_b - \bar{w}_b^*| \le \epsilon_w, \qquad |\widehat{e}_b - \bar{e}_b^*| \le \epsilon_e. \tag{5}$$

For order r, let

$$g_r(w, e) = p_r \max_{0 \le u \le 1} \{u F_r(u, w, e \mid W_r, E_r)\} \tag{6}$$

denote its optimized expected revenue contribution. Let L_w and L_e be finite constants that bound the sensitivity of this optimized expected revenue to pickup time and detour ratio over the relevant range, so that

$$|g_r(w, e) - g_r(w', e')| \le p_r \left(L_w |w - w'| + L_e |e - e'|\right). \tag{7}$$

The predictor-induced pattern-valuation error is then bounded by

$$B_b^{\text{pred}} = C\epsilon_\ell + \sum_{r \in S_b} p_r \left(L_w \epsilon_w + L_e \epsilon_e\right), \tag{8}$$

and the triangle inequality gives

$$|v_b - \widehat{a}_b| \le A_b + B_b^{\text{pred}}. \tag{9}$$

Third, if the A-CAM proxy master is not solved to exact optimality, let Γ denote any valid absolute residual gap satisfying

$$\max_{x \in \mathcal{X}} \widehat{V}(x) - \widehat{V}(\widehat{x}) \le \Gamma, \tag{10}$$

where

$$\widehat{V}(x) = \sum_{b \in \mathcal{B}} \widehat{a}_b x_b. \tag{11}$$

For any feasible selection $x \in \mathcal{X}$, define

$$E(x) = \sum_{b \in \mathcal{B}} \left(A_b + B_b^{\text{pred}}\right) x_b. \tag{12}$$

Equation (9) immediately implies

$$|V(x) - \widehat{V}(x)| \le E(x), \qquad \forall x \in \mathcal{X}. \tag{13}$$

1.2 Conditional Downstream-Loss Bound

Theorem 4 (Conditional downstream-loss bound)**.** *Under the predictor-error bounds in* (5)*,*

$$Z^* - V(\widehat{x}) \le \Gamma + E(x^*) + E(\widehat{x}) \le \Gamma + 2 \max_{x \in \mathcal{X}} E(x). \tag{14}$$

Proof. Adding and subtracting the proxy values at x^* and $\widehat{x}$ gives

$$Z^* - V(\widehat{x}) = \underbrace{V(x^*) - \widehat{V}(x^*)}_{\text{valuation error at } x^*} + \underbrace{\widehat{V}(x^*) - \widehat{V}(\widehat{x})}_{\text{proxy-master solution gap}} + \underbrace{\widehat{V}(\widehat{x}) - V(\widehat{x})}_{\text{valuation error at } \widehat{x}}. \tag{15}$$

By (13), the first and third terms are bounded above by $E(x^*)$ and $E(\widehat{x})$, respectively. By (10), the middle term is bounded above by Γ. Therefore,

$$Z^* - V(\widehat{x}) \le \Gamma + E(x^*) + E(\widehat{x}).$$

Finally,

$$E(x^*) + E(\widehat{x}) \le 2 \max_{x \in \mathcal{X}} E(x),$$

which yields the second inequality in (14). □

Theorem 4 does not require the discrete matching decision to remain unchanged under predictor perturbations. The term $E(x^*)$ controls possible undervaluation of the exact optimum, whereas $E(\widehat{x})$ controls possible overvaluation of the matching selected by A-CAM. Therefore, ranking reversals and matching switches are explicitly accommodated.

1.3 Problem-Size Dependence and Predictor Calibration

Suppose an instance contains n orders and K vehicles, each vehicle can serve at most N orders, and $p_r \le p_{\max}$. Any feasible matching can contain at most

$$M = \min\{n, K\} \tag{16}$$

nonempty vehicle patterns and serve at most

$$J = \min\{n, NK\} \tag{17}$$

orders. Define

$$A_{\max} = \max_{x \in \mathcal{X}} \sum_{b \in \mathcal{B}} A_b x_b. \tag{18}$$

For any feasible x,

$$E(x) \le A_{\max} + CM\epsilon_\ell + p_{\max} J \left(L_w \epsilon_w + L_e \epsilon_e\right). \tag{19}$$

Substituting (19) into Theorem 4 yields

$$Z^* - V(\widehat{x}) \le \Gamma + 2A_{\max} + 2\left[CM\epsilon_\ell + p_{\max} J \left(L_w \epsilon_w + L_e \epsilon_e\right)\right]. \tag{20}$$

Thus, for fixed predictor-error radii and model parameters, the predictor-induced component grows at most linearly with the maximum numbers of active vehicles and served orders.

The statistical calibration of predictor errors and the deterministic bound play different roles. Split-conformal calibration provides statistically meaningful scales for ϵ_ℓ, ϵ_w, and ϵ_e under the corresponding exchangeability conditions, but it does not by itself bound downstream optimization loss. Theorem 4 instead characterizes what follows conditionally when the relevant pattern-level prediction errors satisfy the specified bounds. The numerical audit uses the common base radii

$$\epsilon_\ell = 2.4988, \qquad \epsilon_w = 7.4444, \qquad \epsilon_e = 0.0949, \tag{21}$$

all reported to four decimal places. The tour-length radius 2.4988 is the value after the unit conversion used in the implementation.

1.4 Exact Worst-Case Inducible Matching

The downstream worst case is determined by the matching induced by predictor perturbations rather than solely by the absolute deterioration of the proxy objective. Simultaneously worsening all predicted service attributes can reduce many proxy values together while leaving their relative ranking unchanged. By contrast, asymmetric pattern-level perturbations can change the ranking and induce a matching with a substantially lower exact executable value.

For the numerical analysis, the predictor-error magnitude is controlled by

$$\alpha \in \{0, 0.25, 0.5, 1, 1.5, 2\}. \tag{22}$$

For each pattern b, the admissible predictor attributes satisfy

$$\begin{aligned} \widehat{\ell}_b &\in [\max\{0, \ell_b^* - \alpha\epsilon_\ell\},\ \ell_b^* + \alpha\epsilon_\ell], \\ \widehat{w}_b &\in [\max\{0, \bar{w}_b^* - \alpha\epsilon_w\},\ \bar{w}_b^* + \alpha\epsilon_w], \\ \widehat{e}_b &\in [\max\{0, \bar{e}_b^* - \alpha\epsilon_e\},\ \bar{e}_b^* + \alpha\epsilon_e]. \end{aligned} \tag{23}$$

The exact worst-case audit adopts a patternwise Cartesian box, so the admissible predictor realization for one pattern can be selected independently of the others.

Let $\underline{a}_b(\alpha)$ and $\overline{a}_b(\alpha)$ denote the minimum and maximum attainable A-CAM values of pattern b over its admissible predictor box. For a candidate matching $x \in \mathcal{X}$, define the candidate-favoring coefficient vector

$$a_b^x(\alpha) = \begin{cases} \overline{a}_b(\alpha), & x_b = 1, \\ \underline{a}_b(\alpha), & x_b = 0. \end{cases} \tag{24}$$

This vector gives selected patterns their largest attainable A-CAM values and unselected patterns their smallest attainable values.

Lemma 1 (Exact inducibility test)**.** *Under the patternwise Cartesian coefficient box, a feasible matching $x \in \mathcal{X}$ can be induced as an A-CAM-optimal response by at least one admissible predictor realization if and only if x is A-CAM-optimal under the candidate-favoring vector $a^x(\alpha)$.*

Proof. For any competitor $y \in \mathcal{X}$, define the objective margin of x relative to y as

$$\Delta_{x,y}(a) = \sum_{b \in \mathcal{B}} a_b(x_b - y_b). \tag{25}$$

If $x_b - y_b = 1$, this margin is maximized by setting $a_b = \overline{a}_b(\alpha)$; if $x_b - y_b = -1$, it is maximized by setting $a_b = \underline{a}_b(\alpha)$; and if $x_b - y_b = 0$, the coefficient does not affect the margin. Hence, the single vector $a^x(\alpha)$ maximizes $\Delta_{x,y}(a)$ simultaneously for every competitor y.

If x is not optimal under $a^x(\alpha)$, some competitor y satisfies $\Delta_{x,y}(a^x(\alpha)) < 0$. Since $a^x(\alpha)$ maximizes this margin, the same competitor dominates x under every admissible coefficient realization, so x is not inducible. Conversely, if x is optimal under $a^x(\alpha)$, then $a^x(\alpha)$ itself is an admissible witness under which x is A-CAM-optimal. □

For instance i, define the inducible matching set

$$\mathcal{X}_i^{\text{ind}}(\alpha) = \left\{ x \in \mathcal{X}_i : x \in \arg\max_{y \in \mathcal{X}_i} \widehat{V}(y; \delta) \text{ for some } \delta \in \mathcal{U}_\alpha \right\}. \tag{26}$$

Proposition 1 (Worst-case inducible matching)**.** *For instance i and error multiplier α, the worst-case downstream loss over the prescribed predictor-error set is*

$$L_i^{\mathrm{wc}}(\alpha) = Z_i^* - \min_{x \in \mathcal{X}_i^{\mathrm{ind}}(\alpha)} V(x). \tag{27}$$

Consequently, if feasible matchings are examined in nondecreasing order of their exact executable value $V(x)$, the first inducible matching is a globally worst A-CAM response over the prescribed predictor-error set.

Proof. For a fixed instance, Z_i^* is constant. Maximizing $Z_i^* - V(x)$ over the inducible responses is therefore equivalent to minimizing $V(x)$ over $\mathcal{X}_i^{\mathrm{ind}}(\alpha)$. The ordering statement follows immediately. □

Because the predictor-error sets are nested as α increases, the globally worst inducible loss is nondecreasing in α whenever the exact search completes without truncation.

1.5 Exact Worst-Case Search Procedure

Proposition 1 gives a direct search strategy. For each instance, enumerate

$$\mathcal{X}_i = \{x^{(1)}, \ldots, x^{(N_i)}\}$$

and sort the matchings so that

$$V(x^{(1)}) \leq \cdots \leq V(x^{(N_i)}).$$

Starting from the matching with the lowest exact executable value, apply Lemma 1 to test whether the candidate can be induced as an A-CAM-optimal response. The first inducible matching is globally worst when the search terminates without truncation.

Algorithm 1 Exact worst-case predictor-error sensitivity analysis

Require: Instance set $\mathcal{I}$; error multipliers $\{0, 0.25, 0.5, 1, 1.5, 2\}$; base radii $(\epsilon_\ell, \epsilon_w, \epsilon_e)$; absolute optimality tolerance τ; optional time limit T

Ensure: Worst-case inducible matching, exact downstream loss, and theoretical bounds

for all $i \in \mathcal{I}$ **do**
  Load the complete feasible pattern set $\mathcal{B}_i$ and, for each pattern, $(\ell_b^*, \bar{w}_b^*, \bar{e}_b^*)$, v_b, and A_b
  Solve the exact pattern master to obtain x_i^* and $Z_i^* = V(x_i^*)$
  Enumerate $\mathcal{X}_i = \{x^{(1)}, \ldots, x^{(N_i)}\}$ and sort by nondecreasing $V(x)$
  for all $\alpha \in \{0, 0.25, 0.5, 1, 1.5, 2\}$ **do**
    Compute $[\underline{a}_b(\alpha), \overline{a}_b(\alpha)]$ for every $b \in \mathcal{B}_i$
    Initialize a known inducible incumbent $\widehat{x}_i^{\mathrm{wc}}$ and set complete $\leftarrow$ false
    for $q = 1, \ldots, N_i$ **do**
      if time limit T is reached **then**
        break
      end if
      Set $x \leftarrow x^{(q)}$ and construct $a^x(\alpha)$ from (24)
      Solve the A-CAM best-response master under $a^x(\alpha)$
      if x is A-CAM-optimal within the prescribed absolute tolerance τ **then**
        $\widehat{x}_i^{\mathrm{wc}} \leftarrow x$; complete $\leftarrow$ true
        break
      end if
    end for
    if complete = true **then**
      Certify $\widehat{x}_i^{\mathrm{wc}}$ as globally worst over the prescribed predictor-error set
    else
      Retain the best available inducible incumbent and record incomplete-search status
    end if
    Construct the endpoint witness and independently verify A-CAM optimality within τ
    Compute $L_i^{\mathrm{wc}}(\alpha) \leftarrow Z_i^* - V(\widehat{x}_i^{\mathrm{wc}})$
    Compute the selected-solution and uniform theoretical bounds
    Record the matching, loss, bounds, search status, candidate count, and runtime
  end for
end for

The optional time limit is an implementation safeguard. Global-worst-case language is used only when the search completes without truncation; otherwise, the retained matching is treated as the best available inducible incumbent and the search status is reported explicitly.

For error multiplier α, define the scaled pattern-level error allowance

$$D_b(\alpha) = A_b + \alpha \left[C\epsilon_\ell + \sum_{r \in S_b} p_r \left(L_w \epsilon_w + L_e \epsilon_e \right) \right] \tag{28}$$

and

$$E_\alpha(x) = \sum_{b \in \mathcal{B}_i} D_b(\alpha) x_b. \tag{29}$$

Let τ denote the prescribed absolute objective-value tolerance in the best-response optimality test.

For a certified best-response solve, the residual proxy-master gap satisfies $\Gamma \leq \tau$. The selected-solution bound used in the numerical audit is therefore

$$B_i^{\mathrm{sel}}(\alpha) = \tau + E_\alpha(x_i^*) + E_\alpha(\widehat{x}_i^{\mathrm{wc}}), \tag{30}$$

while the corresponding uniform form is

$$B_i^{\mathrm{unif}}(\alpha) = \tau + 2 \max_{x \in \mathcal{X}_i} E_\alpha(x). \tag{31}$$

The numerical audit checks

$$L_i^{\mathrm{wc}}(\alpha) \leq B_i^{\mathrm{sel}}(\alpha) \leq B_i^{\mathrm{unif}}(\alpha). \tag{32}$$

1.6 Implementation Verification and Numerical Results

The worst-case search uses perturbed A-CAM values only to determine which matching is induced. Downstream performance is evaluated using precomputed exact fixed-pattern executable profits. In the implementation, the fixed-pattern cache is generated by `generate_exact_pattern_cache.py`. For each feasible pattern, the corresponding fixed-pattern joint O-CAM problem reoptimizes the vehicle route, continuous pricing, pickup-time and detour-related service attributes, and acceptance probabilities. Only a proven optimal objective value is retained as v_b.

The induced matching x is therefore evaluated as

$$V_{\mathrm{cache}}(x) = \sum_{b \in \mathcal{B}_i} v_b x_b, \tag{33}$$

and the implementation-level worst-case loss is

$$L_{i,\mathrm{impl}}^{\mathrm{wc}}(\alpha) = Z_i^* - \sum_{b \in \mathcal{B}_i} v_b \widehat{x}_{ib}^{\mathrm{wc}}. \tag{34}$$

The search does not rerun Policy 2 after identifying $\widehat{x}_i^{\mathrm{wc}}$; instead, it uses these exact fixed-pattern executable values. The cache model and the Policy 2 route-and-pricing post-optimization share the same core fixed-matching objective and decision structure. As an implementation check, 480 fixed-pattern cases were cross-checked between the cache values and the direct post-optimization calculations. All corresponding subproblems were solved to optimality, and the objective values agreed within the prescribed numerical tolerance.

The numerical audit uses ten fixed instances from each of A_{4_2}, A_{6_3}, and A_{8_4}, yielding 30 instances in total. Each instance is evaluated at six predictor-error multipliers in (22), giving 180 instance–error-level pairs. At error multiplier α, the three base radii in (21) are multiplied by α.

As shown in Fig. 1, the selected-solution theoretical bound covers the identified worst-case downstream loss for every tested instance and every error level. At $\alpha = 0$, the predictor-induced component vanishes, while the bound can remain positive because the structural aggregation error A_b is still present. This is consistent with the decomposition in Theorem 4, which separates structural aggregation error from predictor-induced valuation error.

As α increases, the admissible predictor-error set expands and the worst-case downstream loss generally increases. For globally completed searches, the nested uncertainty sets imply that the exact worst-case loss is nondecreasing in α. Larger instances also tend to exhibit larger absolute

Table 1: Design of the exact worst-case predictor-error audit

Item	Setting
Instance groups	A_{4_2}, A_{6_3}, A_{8_4}
Instances per group	10
Total instances	30
Error multipliers	$\{0, 0.25, 0.5, 1, 1.5, 2\}$
Instance–error-level pairs	180
Base predictor-error radii	(2.4988, 7.4444, 0.0949)
Downstream evaluation	Exact fixed-pattern executable values
Primary diagnostic	$B_i^{\mathrm{sel}}(\alpha)$

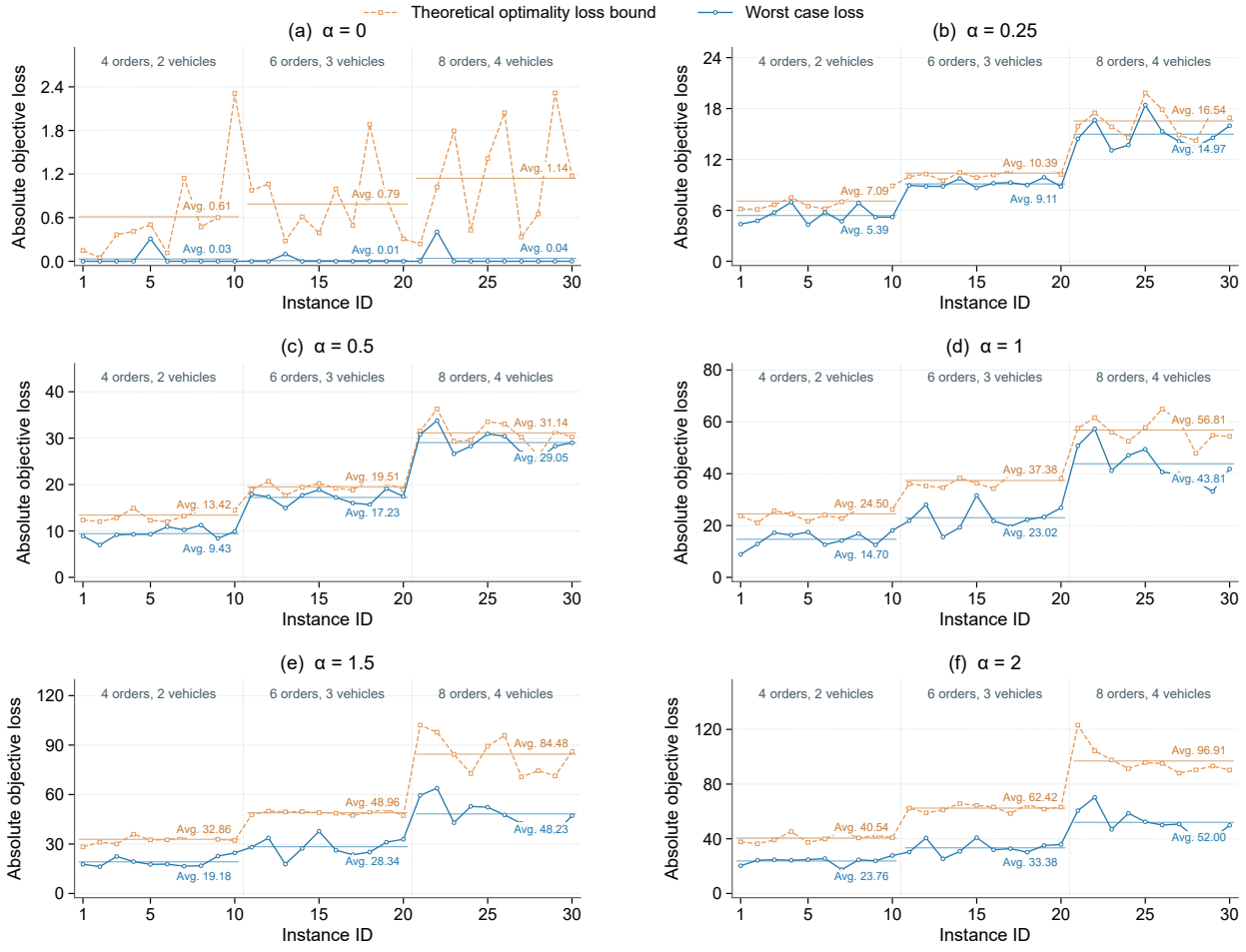


Figure 1: Worst-case downstream optimization loss and selected-solution theoretical bound under bounded predictor errors. Panels (a)–(f) correspond to $\alpha \in \{0, 0.25, 0.5, 1, 1.5, 2\}$. Instance IDs 1–10, 11–20, and 21–30 correspond to A_{4_2}, A_{6_3}, and A_{8_4}, respectively. The blue curve reports the identified worst-case downstream loss, and the orange curve reports the corresponding selected-solution theoretical bound.

worst-case losses and bounds, which is consistent with the explicit problem-size dependence in (20). This numerical pattern should be interpreted as consistency with the theoretical size dependence rather than as an empirical proof of linear scaling.

The distance between the realized worst-case loss and the selected-solution bound varies across instances and error levels. At moderate error levels, the identified worst-case loss is relatively close to the bound in many cases. At larger error levels, the theoretical allowance grows continuously with the admissible predictor radius, whereas the realized loss changes through discrete ranking reversals and matching switches; the bound can therefore become more conservative while still covering the identified worst-case loss. Overall, the experiment provides a direct run-by-run numerical audit of the conditional downstream-loss bound under adverse inducible matching responses.